

23.09.26

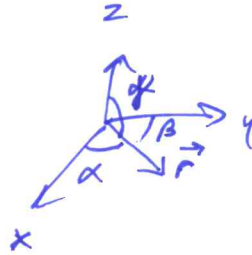
PHYS 325

1. Ders

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• Vectors in Cartesian Coordinate System:

$$\vec{r} \leftrightarrow (x, y, z)$$



R.H: Right Hand.

α : Angle b/w x axis and $\vec{r}$

β : " y "

γ : " z "

$$x: r \cdot \cos \alpha$$

$$y: r \cdot \cos \beta$$

$$z: r \cdot \cos \gamma$$

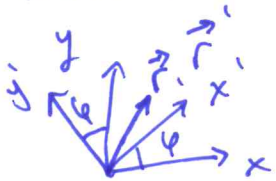
direction
cosines.

$A_{x,y,z}$: cartesian coords of $\vec{A}$ vector.

$\vec{A} = A_x \cdot \hat{x} + A_y \cdot \hat{y} + A_z \cdot \hat{z}$ where $\hat{x}, \hat{y}, \hat{z}$ are the unit vectors.

$$|\vec{A}| = (A_x^2 + A_y^2 + A_z^2)^{\frac{1}{2}}$$

Rotation of coordinate axis



$$x = r \cdot \cos Q$$

$$Q' = Q - \phi$$

$$y = r \cdot \sin Q$$

$$x' = r \cdot \cos Q' = r \cdot \cos(Q - \phi) = r [\sin Q \sin \phi + \cos Q \cos \phi]$$

$$y' = r \cdot \sin Q'$$

$$= (r \sin Q) \sin \phi + (r \cos Q) \cos \phi$$

$$y' = (r \sin Q) \cos \phi - (r \cos Q) \sin \phi$$

$$x' = \cos \phi \cdot x + \sin \phi \cdot y$$

$$y' = -\sin \phi \cdot x + \cos \phi \cdot y$$

(1)

(x', y') depends on (x, y) and transformation parameter φ .

$$\begin{aligned} x &\rightarrow x_1 \\ y &\rightarrow x_2 \\ \text{and} \\ z &\rightarrow x_3 \\ &\vdots \end{aligned} \quad \begin{aligned} x'_1 &= a_{11}x_1 + a_{12}x_2 \\ x'_2 &= a_{21}x_1 + a_{22}x_2 \end{aligned} \quad \left| \begin{aligned} \text{with } a_{11} &= \cos \varphi, a_{12} = \sin \varphi \\ a_{21} &= -\sin \varphi, a_{22} = \cos \varphi \end{aligned} \right.$$

then we have a notation

$$x'_i = \sum_{j=1}^N a_{ij} \cdot x_j, \quad \begin{matrix} i=1 \\ i=2 \\ \vdots \\ i=N \end{matrix}$$

Generalized Transformation.

2. Ders

transformation: $x'_i = \sum_{j=1}^N a_{ij} x_j, \quad i=1, \dots, N.$

$$\frac{\partial x'_i}{\partial x_j} = \frac{\partial}{\partial x_j} [a_{i1} \cdot x_1 + \dots + \underbrace{a_{ij} x_j}_{\text{term}} + \dots + a_{iN} \cdot x_N] = a_{ij}$$

Inverse transformation for rotation. $\varphi \rightarrow -\varphi$

$$x_j = \sum_{i=1}^N b_{ji} \cdot x'_i \quad \text{with } b_{ji} = \frac{\partial x_j}{\partial x'_i}$$

Ex For rotation transformation show that $b_{ji} = a_{ij}$, in other words

$$\frac{\partial x_j}{\partial x'_i} = \frac{\partial x_j}{\partial x'_i}$$

$$x_i' = \sum_{j=1}^N \underbrace{\frac{\partial x_i'}{\partial x_j}}_{a_{ij}} \cdot x_j = \sum_{j=1}^N \frac{\partial x_j}{\partial x_i} \cdot x_j$$

Note that direction cosines a_{ij} satisfy orthogonality condition.

$$\sum_{i=1}^N a_{ij} \cdot a_{ik} = \delta_{jk}$$

or

$$\sum_{j=1}^N a_{ji} \cdot a_{ki} = \delta_{jk} \text{ where } \delta_{jk} \text{ is}$$

Kronecker delta defined as

$$\delta_{jk} = \begin{cases} 1 & j=k \\ 0 & \text{otherwise} \end{cases}$$

$$\sum_{i=1}^N \frac{\partial x_j}{\partial x_i'} \cdot \frac{\partial x_k}{\partial x_i'} = \sum_i \frac{\partial x_j}{\partial x_i'} \cdot \frac{\partial x_i'}{\partial x_k} = \frac{\partial x_j}{\partial x_k} = \delta_{jk}$$



Homework.

Confirmation.

$$\begin{aligned} x_1' &= \cos \varphi x_1 + \sin \varphi x_2 \\ x_2' &= -\sin \varphi x_1 + \cos \varphi x_2 \end{aligned}$$

$$\begin{aligned} x_1 &= \cos \varphi x_1' - \sin \varphi x_2' \\ x_2 &= \sin \varphi x_1' + \cos \varphi x_2' \end{aligned}$$

$$x_i' = \sum_{j=1}^2 a_{ij} x_j \rightarrow i=1$$

$$x_1' = \sum_j a_{1j} x_j = a_{11} x_1 + a_{12} x_2$$

$$i=2 \quad x_2' = \sum_j a_{2j} x_j = a_{21} x_1 + a_{22} x_2$$

$$\delta_{jk} \stackrel{?}{=} \sum_i a_{ij} \cdot a_{ik}$$

$$j=k=1 \Rightarrow \delta_{11} \stackrel{?}{=} 1 = \sum_i a_{i1} \cdot a_{i1}$$

$$= a_{11} \cdot a_{11} + a_{21} \cdot a_{21}$$

$$(\cos \varphi)^2 + (-\sin \varphi)^2$$

$$= 1.$$

$$j=1, k=2 \quad \delta_{12} = 0 = a_{11} a_{12} + a_{21} \cdot a_{22}$$

$$\cos \varphi \sin \varphi - \cos \varphi \cdot \sin \varphi$$

$$= 0.$$

$$\boxed{(\mathbf{I})_{ij} = \delta_{ij}}$$

Scalar Product

~~Physically we need this ~~quantity~~ quantity~~

Physically we need this scalar quantity which depends on the $\vec{A}, \vec{B}$ vector

1) Proportional to $|\vec{A}|$

2) " $|\vec{B}|$

3) " cosine between $\vec{A}$ and $\vec{B}$.

Example: Work.

$$\boxed{W = |\vec{F}| \cdot |\Delta \vec{r}| \cdot \cos \theta}$$

Then this operation

$$\vec{A} \cdot \vec{B} = |\vec{A}| \cdot |\vec{B}| \cdot \cos \alpha$$

In three-d space: following special cases

$$\vec{A} \cdot \hat{x} = A \cdot 1 \cdot \cos \alpha = A_x$$

$$\vec{A} \cdot \hat{y} = A_y$$

$$\vec{A} \cdot \hat{z} = A_z.$$

$$\vec{A} \cdot \vec{B} = \vec{A} \cdot (B_x \hat{x} + B_y \hat{y} + B_z \hat{z})$$

$$= B_x (\vec{A} \cdot \hat{x}) + B_y (\vec{A} \cdot \hat{y}) + B_z (\vec{A} \cdot \hat{z})$$

$$\vec{A} \cdot \vec{B} = A_x B_x + B_y A_y + A_z B_z$$

$$\vec{A} \cdot \vec{B} = \sum_i A_i B_i \Rightarrow \vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

$$\left| \begin{array}{l} \hat{x} \cdot \hat{x} = 1 \\ \hat{y} \cdot \hat{y} = 1 \\ \hat{z} \cdot \hat{z} = 1 \end{array} \right.$$

$$\rightarrow \hat{x} \cdot \hat{y} = \hat{y} \cdot \hat{z} = \hat{z} \cdot \hat{x} = 0$$

$$\rightarrow \hat{x}_i \cdot \hat{x}_j = \delta_{ij}$$

(Notation: 1) latin letter indexes 1... 3

greek indices 0, 1, 2, 3

$$\vec{A} \cdot \vec{B} = \sum_i A_i B_i = A_i \cdot B_i$$

↑
implied summation.

• invariant or scalar product